Botho: Privacy-Preserving Cryptocurrency

Executive Summary

Overview

Botho is a privacy-preserving cryptocurrency that combines post-quantum cryptography with efficient transaction privacy, achieving strong security guarantees with practical performance.

Key innovation: Hybrid post-quantum architecture applies quantum-resistant cryptography selectively—protecting permanent data (recipient identities) while using efficient classical cryptography for ephemeral privacy (sender anonymity).

Core Features

Feature	Description
Privacy	All transactions private by default. Ring signatures (CLSAG) hide sender among 20 decoys. Stealth addresses hide recipients. Confidential transactions hide amounts.
Post-Quantum	ML-KEM-768 protects recipient identity against future quantum attacks. On-chain data remains secure even if quantum computers break classical cryptography.
Fast Finality	Hybrid PoW+SCP consensus achieves deterministic finality in ∼5 seconds. Unlike Bitcoin, finalized blocks cannot be reverted.
Fair Economics	Progressive fees and demurrage based on coin ancestry (not identity) are aimed at wealth concentration. Fees are 80% paid out via lottery, 20% burned. A measurable reduction in inequality is *not* demonstrated at production parameters (see Economic Design).
Cross-chain Bridge	Native BTH wraps 1:1 to transparent **wBTH** on Ethereum and Solana (factor-1 peg; a t-of-n validator federation custodies the reserve). Unwrapped coins enter tagged to a block-epoch *import cluster* (epoch $K = 1$ day, floor $F = 1.5\times$), so freshly-imported wealth is expensive until it circulates—closing the lineage-reset round-trip.

Technical Parameters

Parameter	Value	Notes
Ring size	20	Sender hidden among 20 possibilities
Transaction size	∼4 KB	Practical for mobile wallets
Block time	3–40s	Dynamic based on load (5s monetary reference)
Finality	∼5s	Deterministic (not probabilistic)
Tail emission	2.5% gross/yr	Supply-dependent (∼2.42 BTH/block at 611M)

Cryptographic Primitives

- **Recipient privacy**: ML-KEM-768 (post-quantum KEM, NIST FIPS 203)
- **Sender privacy**: CLSAG ring signatures (classical, efficient)
- **Amount privacy**: Pedersen commitments + Bulletproofs range proofs
- **Minting attribution**: PoW preimage binding (hash-based; quantum-resistant, no signature)
- **Key derivation**: BIP39 mnemonic → SLIP-10 → subaddresses

Consensus Mechanism

Hybrid PoW + SCP: Proof-of-work provides permissionless block proposal (anyone can mine). Stellar Consensus Protocol provides fast Byzantine-fault-tolerant finalization.

Why hybrid?

- PoW alone: Slow probabilistic finality, reorg vulnerability
- BFT alone: Requires known validator set, not permissionless
- PoW+SCP: Permissionless *and* fast deterministic finality

Economic Design

Progressive fees: Transaction fees scale with sender's cluster wealth percentile (1× for bottom 50%, up to 6× for top 1%). Based on coin *ancestry*, not current ownership—preserves privacy.

Fee distribution:

- 80% redistributed via a lottery to 4 winning UTXOs per block, drawn uniformly from recently created outputs (with a PoW-bound seed); its anti-splitting cap does not bind as built (#1682)
- 20% burned (creates deflationary pressure proportional to usage)

Redistribution status: the mechanisms are specified and implemented as described, but measurable inequality reduction is *not demonstrated* at production parameters. A code-faithful simulation (#1790, #1808) measures the mechanism's effect on the Gini coefficient at -0.028 to $+0.003$ against a 0.05 design bar, and even at 1000× the transaction rate no case clears the bar at both 10 and 20 years. The historical stationary splitting model found profitable tickets (#1761, #1682); current quantitative production profitability needs a production-mode re-run. A sim-only redesign is under study (#1801); no constant has changed.

Tail emission: After 5 annual halvings (∼5 years, fixing the Phase-1 supply at ≈611M BTH), perpetual tail emission funds long-term security at a 2.5% gross annual rate (designed as 2% net of a 0.5% fee burn; at realistic fee volumes the burn is much smaller, so net inflation is close to 2.5%). This is supply-dependent (≈2.42 BTH/block gross at the 611M tail-onset supply, growing with supply), not a fixed per-block constant.

Security Model

Threat	Protection	Security Level
Sender identification	CLSAG ring signatures	128-bit classical
Recipient identification	ML-KEM-768 stealth addresses	192-bit post-quantum
Amount discovery	Pedersen + Bulletproofs	128-bit classical
Double-spending	Key images + SCP finality	Information-theoretic
51% attack	SCP quorum intersection	Byzantine fault tolerant

Comparison

	Botho	Monero	Zcash	Bitcoin
Privacy	Full	Full	Optional	Pseudonymous
Post-quantum	Partial	No	No	No
Finality	5s deterministic	10+ min prob.	10+ min prob.	60+ min prob.
Tx size	4 KB	2 KB	2 KB (shielded)	0.3 KB
Progressive fee schedule	Yes	No	No	No

Resources

- **Full whitepaper**: `botho-whitepaper.pdf` (129 pages)
- **Website**: `https://botho.org`
- **Source code**: `https://github.com/botho/botho`

"Motho ke motho ka batho" — A person is a person through other people.